Tribhuvan University

Faculty of Science and Technology

OFFICE OF THE DEAN

2082

Bachelor of Science in Computer Science and Information Technology

Course Title: **Computer Architecture**

Code No: **CSC165**   Semester: **3rd Semester**

Full Marks: **60**   Pass Marks: **24**   Time: **3 hours**

Candidates are required to answer the questions in their own words as far as practicable.

Section A

1. Explain pipelining processing with an example of your own. Illustrate pipelining speed up equations under various conditions.

2. Differentiate between restoring and non-restoring division algorithm. Explain the restoring division algorithm with an example of your own.

3. Describe the interrupt cycle. Explain the interrupt cycle with the required flowchart.

Section B

Attempt any Eight questions[8×5=40]

4. List different arithmetic microorganisms. Explain 4-bit binary adder-subtractor.

5. Briefly explain memory reference instructions.

6. List different binary codes. Explain any two of the binary codes.

7. Explain any five addressing modes with an example of each.

8. Define vector operation. Explain the concept of vector operation with an example of matrix multiplication.

9. Illustrate the working mechanism of Direct Memory Access in association with the concept of cycle stealing. How DMA is different from IOP.

10. Explain microprogram sequencer for control memory with the required figure.

11. Describe temporal locality and spatial locality. Differentiate between associative mapping and set associative mapping in cache memory.

12. Write short notes ona) Handshakingb) Control word